

Capstone Proposal: Curico Classroom

1. Cover / Header

Topic title	Curico Classroom: แพลตฟอร์ม AI สนับสนุนการจัดกิจกรรม Hands-on และติดตามความเข้าใจของนักเรียนรายบุคคล
Topic ID / code	72
Prepared for	Curico AI
Team name / number	BMCC
Team members	1. Piyongkul Rardyota (coordinator) · ID 6632134521 · 6632134521@student.chula.ac.th · (+66) 61-556-2770 2. Rachata Kanitpunyacharoen · rachatakanit@gmail.com 3. Chayakorn Unhasirikul · chayakorn26082547@gmail.com 4. Ratsaporn Prathumma · rodolf8680@gmail.com
Prepared date	August 14, 2026
Round	3

2. Problem Understanding

Curico AI builds tools for hands-on education. Hands-on activities give better learning results than lectures alone. But hands-on classes are difficult to operate. In a class of 30–40 students, one teacher cannot see the progress of each student. Each student works at a different speed. Questions, errors, and misconceptions occur at many desks at the same time. Most of this information is lost. The teacher gets no record of who was confused, about what, and when.

One-to-one tutoring gives large learning gains. Schools cannot give each student a human tutor. Computer tutors that guide students step by step come near the effect of human tutors. Large language models (LLMs) now make this guidance possible in natural conversation.

The topic asks for one platform with six parts: classroom management for teachers; a step-by-step activity interface for students; an AI assistant that guides students with hints, not with direct answers; a dashboard that shows the teacher which students need help; AI-assisted formative assessment that drafts feedback for the teacher; and safety, privacy, and teacher-control functions. The teacher makes all final decisions. Curico AI also wants evidence from the prototype: how much teacher load the AI removes, which data helps teachers decide, and which alerts and summaries are useful. Curico AI plans to grow good results into product features for schools.

3. Assumptions

1. **Pilot scale.** We assume a pilot with 1–3 classrooms, up to 40 students per classroom, and up to 150 concurrent users. A standard three-tier web architecture is sufficient at this scale.
2. **Target users.** We assume secondary-school science or STEM classes as the first use case. The activity structure is generic, so other subjects can use it later.
3. **Devices.** Students use shared tablets, phones, or notebook computers with a browser. School networks can be slow. A Progressive Web Application (PWA) with a small payload and retry logic covers this condition.
4. **Activity content.** Curico AI and teachers supply the activity content (steps, materials, expected results). We build the platform and an activity editor. We do not write curriculum content.
5. **AI model.** We use a commercial LLM through an API. We do not train a foundation model. We control quality with prompt design, retrieval-augmented generation (RAG), and an evaluation process.
6. **AI role.** The AI guides, explains, and drafts. It does not grade and it does not make decisions about a student. This follows the human-centred guidance of UNESCO for generative AI in education and the topic brief.
7. **Voice.** Voice interaction uses a cloud speech-to-text and text-to-speech API. Voice is an input/output option, not a requirement for every task.
8. **Personal data.** We collect only the minimum student data that the functions need. We assume the school or partner organization handles parental or guardian consent for minor students under the privacy rules that apply in their country.
9. **Accounts.** Teachers create classrooms and student accounts or join codes. We assume no integration with an existing school information system.
10. **NDA.** The topic requires a non-disclosure agreement. All four members are ready to sign it.
11. **Success criteria.** We assume these targets: a teacher can find students who need help in less than one minute from the dashboard; AI answers pass a rubric review by teachers; a pilot session with a sample group of teachers or students completes with usability feedback.

4. Proposed Approach / Solution

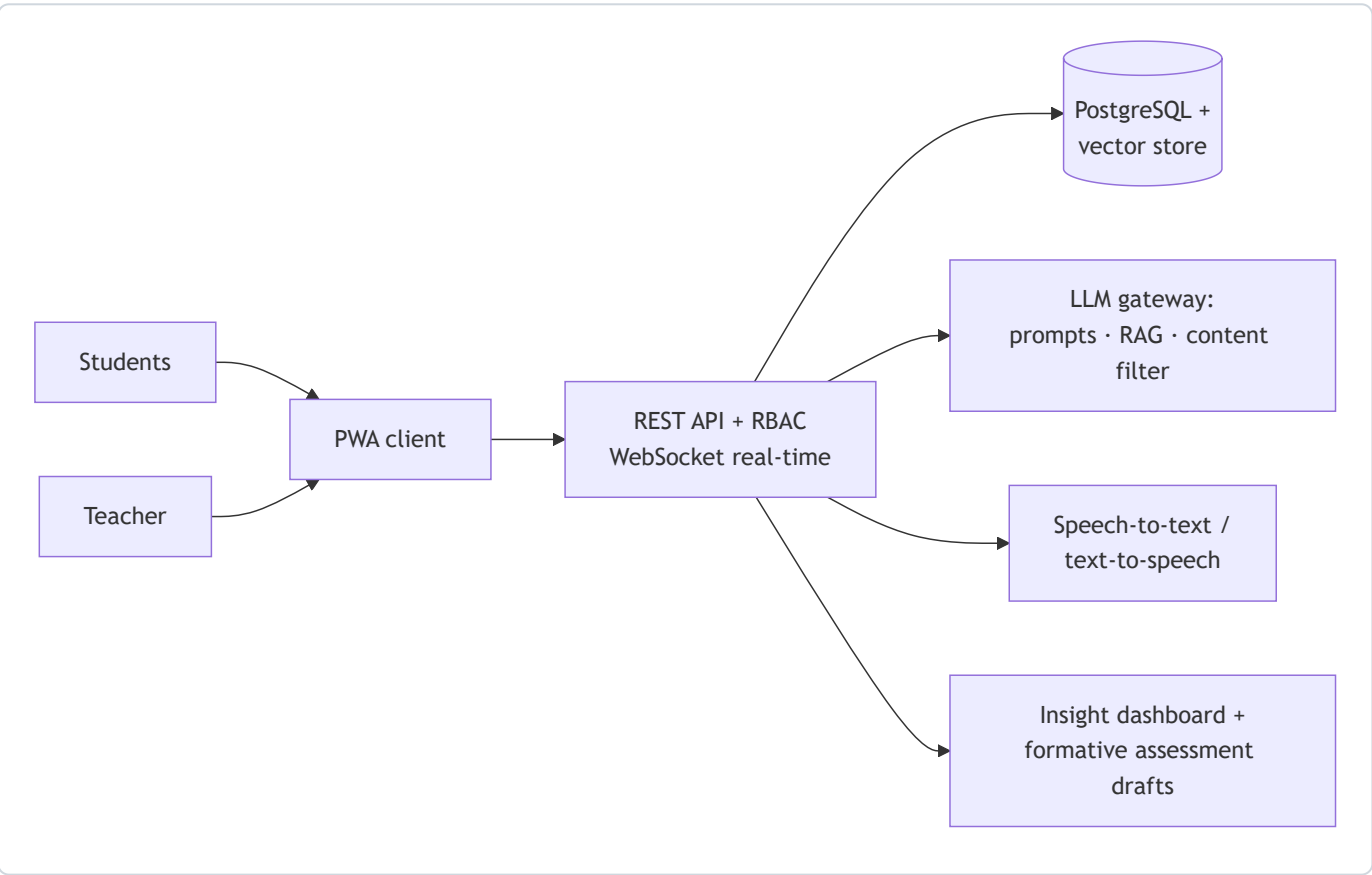
Key idea. The classroom produces learning evidence all the time: answers, questions, chat messages, and photos of experiment results. The platform stores each of these as one structured record in one database. The AI reads these records for two purposes: to guide the student at the desk, and to summarize the class for the teacher. The teacher sees the summary and makes the decision. This is the formative assessment loop, which research shows to be one of the strongest levers on achievement, with AI as the collector and the teacher as the judge.

Main components (mapped to the six scope items in the topic brief)

- **Teacher classroom management.** The teacher creates classrooms, adds students, assigns activities, sets the schedule, follows progress, and sends messages.
- **Student hands-on learning interface.** The student sees the activity steps one at a time. The student records results as text, numbers, choices, or photos. The student asks the AI questions by text or voice.
- **AI learning assistant.** The assistant explains the current step, answers with Socratic hints, and asks questions that make the student think. It does not give the final answer directly. RAG grounds every answer in the teacher's activity sheet, so answers stay on-topic and correct. The assistant flags possible misconceptions and reports them to the teacher, with effective-feedback principles from research.

- **Teacher insight dashboard.** Learning analytics turn the records into views: students who need help now, progress per step, frequent questions, common misconceptions, and per-student and per-class summaries.
- **AI-assisted formative assessment.** The AI reads answers, conversations, and experiment evidence. It drafts feedback and a summary for each student. The teacher reviews, edits, approves, or rejects each draft. The system never releases an automatic judgement.
- **Safety, privacy, and teacher control.** Content filters block unsafe topics. The teacher can turn the AI on or off per activity and can see every AI conversation. Role-based access control (RBAC) limits each role to its own data. Privacy measures cover consent records, data minimization, retention, and deletion.

Architecture. A PWA client connects to a REST API with RBAC. A WebSocket channel pushes real-time progress and alerts to the teacher. Data is in a PostgreSQL database; activity content is embedded in a vector store for RAG. An LLM gateway holds the prompt templates, the content filter, and the cost controls. Speech services convert voice to text and back. The system runs in Docker containers on a cloud host. We follow OWASP guidance for web security.



AI quality process. We keep a golden set of test questions per activity. Each prompt change runs against this set. Teachers score sample answers with a rubric (correct, on-topic, guides without telling, safe). This process is itself a deliverable.

Method. We work in two-week sprints with the Scrum framework.

Phase	Focus
1	Requirement interviews with Curico AI and teachers; UX design and design system
2	Classroom management, activity editor, and the student activity flow

Phase	Focus
3	AI assistant with RAG, voice option, and the AI quality process
4	Dashboard, formative assessment drafts, safety hardening, pilot test, and hand-over

5. Scope & Deliverables

In scope

- Web application / PWA with the six components above.
- Role-based access for teacher, student, and administrator.
- Activity editor and step-by-step student interface with text, number, choice, and photo evidence.
- AI assistant with text and voice interaction, RAG grounding, and Socratic guidance behavior.
- Teacher insight dashboard with per-student and per-class learning analytics.
- AI-drafted, teacher-approved formative feedback.
- Teacher controls: AI on/off per activity, conversation review, and content filters.
- Privacy measures: consent records, data minimization, retention, and delete functions.
- Automated tests and a test report.
- A pilot test with a sample group of teachers or students, and a result report.

Out of scope

- Automatic grading or promotion decisions. The AI drafts; a human decides.
- Curriculum authoring at scale. We supply the editor; Curico AI and teachers supply the content.
- Native mobile applications. The PWA covers mobile use.
- Integration with school information systems or national education platforms.
- Training or fine-tuning of a foundation model.
- Proctoring, identity verification, and payment functions.
- Full offline operation. The PWA tolerates short network drops only.

Expected deliverables / outcomes

- A working prototype on a cloud host, ready for a real classroom trial.
- Source code and an installation guide.
- System architecture and database design documents.
- API documentation.
- UX/UI design and design system.
- AI prompts and the answer-quality evaluation process with results.
- Teacher dashboard and learning analytics views.
- Access control and data protection measures with documentation.
- Automated tests and a test report.
- Pilot test results with a sample group, and recommendations for the path to production.

6. Why This Team

Our four members cover the full stack that this topic needs.

- **Proven delivery as one team.** Three members built and shipped a full-stack reservation platform together (React Native, Express, PostgreSQL) with Agile practice. Team workflow is already in place.
- **AI and LLM engineering.** Piyongkul built LLM pipelines at Cleverse: cost-tiered inference, RAG with pgvector, evaluation harnesses, and prompt-leak defense. This matches the LLM gateway, the RAG design, and the AI quality process. Ratsaporn builds deep-learning models and Thai NLP pipelines (PhayathaiBERT) and designed prompts for a Gemini-based app at LINE HACK 2025.
- **Backend and API experience.** Rachata built gRPC, REST, and GraphQL services at LINE MAN Wongnai. Ratsaporn built microservices with an API gateway, Kafka, and Docker at SiriSoft. This matches the API core and the real-time channel.
- **Frontend experience.** Chayakorn built production web components and role-based UI access at API PLUS TECH. This matches the teacher, student, and dashboard interfaces.
- **Data visualization.** Chayakorn and Ratsaporn built an analytical dashboard for Thai election data. This matches the teacher insight dashboard.
- **Quality and security.** Rachata worked in software quality assurance at CP Axtra (test design, automation with Robot Framework). Piyongkul has applied security engineering (sandboxing, secret management). This supports safe handling of student data and the automated-test deliverable.
- **Cloud and real-time skills.** Piyongkul holds an AWS Academy Cloud Architecting certificate and built sub-0.5-second real-time media systems. Ratsaporn built serverless AWS automation.

All members are fourth-year Computer Engineering students at Chulalongkorn University with strong records in algorithms (three Thailand Olympiad in Informatics medals in the team).

Appendix — Team Member CVs

The CV of each team member follows this page, one page per member, in this order:

1. Piyongkul Rardyota (coordinator)
2. Rachata Kanitpunyacharoen
3. Chayakorn Unhasirikul
4. Ratsaporn Prathumma

Piyongkul Rardyota

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github.com/piyonchu (University) | github.com/potatai (Personal)

Computer Engineering student specializing in **embedded systems, Rust, and AI-powered products built end-to-end**. IOI selection camp qualifier with a track record of optimizing memory-constrained hardware and building sub-0.5s latency platforms.

EDUCATION

Chulalongkorn University <i>Bachelor of Engineering in Computer Engineering</i>	Bangkok, Thailand 2023 – Expected May 2027
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EXPERIENCE

Cleverse <i>Software Engineer Intern</i>	Bangkok, Thailand Jun 2026 – Jul 2026
TaxAI — AI Thai income-tax preparer <i>Rust/Axum, React, Claude + Gemini</i>	
- Designed a cost-tiered extraction pipeline: OpenCV preprocessing (8x fewer vision tokens), cheap Gemini pass, text-only reconciler, Claude escalation — cross-checked by an independent Tesseract OCR witness; 89% amount accuracy on real receipts at ≈\$0.015/doc. - Implemented a deterministic tax engine (the LLM never does the math): 15 deduction categories and 8 progressive brackets, computing returns line by line with Thai Revenue Code citations.	
CanonForge — reference-driven AI game-asset studio <i>Rust/Axum, Postgres + pgvector, ComfyUI</i>	
- Built a versioned style “canon” with an exemplar feedback loop: approved assets are CLIP-embedded and the closest match (blended text + visual cosine similarity) conditions future img2img generation, keeping every asset on-style as the project grows. - Integrated a local generation pipeline on ComfyUI: per-project LoRA training on approved exemplars (kohya + SDXL), ControlNet pose-guided generation, intent-routed inpainting, and single-sheet sprite-animation generation with per-frame auto-crop. - Added pgvector RAG (semantic search, find-similar, dedup, “ask this project” answers) and engine-ready export adapters for Godot and Unity.	
Protected-Skill Marketplace — POC: commercial AI skills that never leak their prompt IP <i>Go, TypeScript, MCP</i>	
- Architected sealed execution: skills run in a network:none kernel namespace behind a deny-all proxy and return only typed results; an AES-256-GCM credential vault and per-run capability tokens keep OAuth secrets out of the sandbox entirely. - Proved the seal adversarially with a layered egress guard (Aho-Corasick shingle matching, per-run canary tokens, LLM semantic judge) against 5 attack families — prompt-extraction rate 0.65 → 0.00 on a held-out corpus.	

PROJECTS

Smart AI IR AC Remote (STM32) <i>C, Picovoice, ESP8266</i>	Link
- Engineered an IR remote by capturing signals and reverse-engineering brand protocols for full AC control, with ESP8266 Wi-Fi + a JavaScript server for remote control and temperature reporting. - Optimized an AI model to run on a tiny STM32F411RE (128KB SRAM / 512KB Flash), enabling on-device wake-word detection and natural-sentence command understanding.	
OmeGroup - Multi-User Video Chat Application <i>Rust, Vite, AWS, Terraform</i>	Link
- Developed a video platform on AWS IVS with sub-0.5s 720p streaming and 7-8x less client RAM than comparable services via efficient WebRTC stream handling and state management. - Architected an event-driven backend (AWS EventBridge signaling, Redis session state, load balancing) with infrastructure fully automated via Terraform.	
FPGA MicroSD GIF Reader & Filter <i>Verilog, FPGA</i>	Link
- Implemented low-level SD card reading and a custom storage format in Verilog to decode GIFs to VGA, sustaining smooth high-FPS playback on 1,800KB RAM by syncing read speed with display refresh. - Built hardware image processing (RGB→HSV grayscale/hue-saturation filters) by simplifying complex formulas into efficient Verilog.	
Additional Projects: <i>Election Analytics Engine, ArtToy Pre-Order (MERN), CP-ZonTana</i>	
Election Analytics: Built a geospatial engine utilizing ML (Isolation Forests) and 3D GPU rendering to audit electoral integrity and map hyper-local voter behavior. CP-ZonTana: Designed a custom, lightweight WebSockets chat protocol in Node.js for real-time bidirectional communication. ArtToy Pre-Order: Full-stack MERN e-commerce platform with JWT auth, injection/XSS hardening, and MongoDB Search Indexes.	

AWARDS & CERTIFICATIONS

Silver Medal & Highest Score (Northeast Region) 18th Thailand Olympiad in Informatics	2022
Qualified for the International Olympiad in Informatics (IOI) selection camp.	
Bronze Medal 1st Thailand Online Mathematics Olympiad (organized by IMO alumni)	2021
AWS Academy Graduate – Cloud Architecting Amazon Web Services	2026
IELTS Academic Overall: 6.5	2023

TECHNICAL SKILLS

Languages: C/C++, Rust, Go, Verilog, Java, TypeScript/JavaScript, Python, SQL, C# Systems & Low-Level: STM32 (ARM Cortex-M), FPGA (Verilog), WebRTC, WebSockets, Reverse Engineering, Memory-Constrained Optimization AI & LLM Systems: Claude/Gemini integration, MCP, RAG (pgvector), prompt caching, cost-tiered inference, eval harnesses Infrastructure: AWS (EC2, IVS, EventBridge), GCP Cloud Run, Terraform, Docker, Redis, Postgres, CI/CD, Linux Frameworks: React, Vite, Axum, Node.js, Express, MongoDB Spoken: Thai (Native), English (IELTS 6.5)	
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RACHATA KANITPUNYACHAROEN

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Wang Mai, Pathum Wan, Bangkok, Thailand, 10330

EDUCATION

Chulalongkorn University
Bachelor of Engineering: Computer Engineering | GPAX 3.14/4.00
• Expected graduation date: May 2027

Bangkok, Thailand
Aug 2023 – Present

SKILLS

Programming: Java, Python, Go, JavaScript/TypeScript, SQL
Frameworks/Tools: Spring, Express.js, PostgreSQL, Docker, Git, Playwright, Robot Framework
Languages: Thai (*native*), English (*intermediate*)

EXPERIENCES

CP Axtra Public Company Limited (Lotus's)
Software Quality Assurance Intern

Bangkok, Thailand
May 2026 – Jul 2026

- Performed manual testing across one physical device app and two web applications, including test case design, evidence capture, and Jira ticket reporting.
- Gained hands-on experience with test automation using Robot Framework (Browser library) by developing a regression test suite for an assigned feature.
- Helped integrate MCP (Model Context Protocol) into the Oracle Retail SIM, enabling AI-assisted test execution and Robot Framework automation via a local server interface.

LINE MAN Wongnai
Software Engineer Intern (Backend)

Bangkok, Thailand
May 2025 – Jul 2025

- Developed gRPC services, REST APIs, and GraphQL endpoints using Java (Spring) and Go to connect microservices and query MySQL data.
- Participated in a Scrum development team, daily stand-ups, and retrospectives to follow the team's workflow and deliver features.

PROJECTS

HewKawJang | Restaurant Reservation Web App
React Native | Express.js | PostgreSQL | Agile/Scrum

Aug 2025 – Nov 2025

- Developed a full-stack reservation platform enabling users to browse restaurants, book time slots, and manage bookings with secure authentication and real-time availability.

Canvas War | Real-Time Multiplayer Pixel-Canvas Battle
HTML/CSS/JS | Express.js | MongoDB

Apr 2024

- Collaborated with classmates to build a r/place-inspired online pixel-art web app where four competing colors fight to claim pixels on a shared canvas in real-time.

ACHIEVEMENTS AND AWARDS

- CU Academic Distinction Award for outstanding contribution in Programming Methodology I (Java and OOP) in the class of 2023.
- Bronze medal in 18th Thailand Olympiad in Informatics (TOI18) in 2022.

EXTRA-CURRICULAR ACTIVITIES

Semi-Finalist | LINE MAN Wongnai Junior Case Competition 2026

Bangkok, Thailand | Mar 2026

- Developed go-to-market strategy with a team of 3 to grow LINE MAN VIP adoption among university students via a company-funded promo code campaign.
- Built the code ledger and automated email distribution pipeline using Google Apps Script.

Team Participant | Thailand Cyber Talent 2025 (Senior)

Bangkok, Thailand | Aug 2025

- Solved challenges in crypto, reverse-engineering, and web exploitation, gaining hands-on cybersecurity experience.

CHAYAKORN UNHASIRIKUL

Bangkok, Thailand | 064-675-2403 | chayakorn26082547@gmail.com

GitHub: github.com/Chayakorn26082547

EDUCATION

Chulalongkorn University

Aug 2023 – May 2027

Bachelor of Engineering in Computer Engineering (GPA: 3.60/4.00)

- Relevant Coursework: Algorithm Design, Data Structures, Software Engineering, Database Systems, Programming Methodology

PROFESSIONAL EXPERIENCE

Frontend Software Engineering Intern

May 2026 – Jul 2026

API PLUS TECH CO., LTD.

- Developed reusable, type-safe frontend components using Svelte 5, TypeScript, and Bits UI for production web applications.
- Integrated GraphQL APIs to support dynamic application features, user workflows, and efficient data retrieval.
- Implemented role-based access control (RBAC) for UI components and application features based on user permissions.
- Migrated legacy frontend components into a modern, maintainable component architecture, reducing technical debt.
- Collaborated closely with frontend and backend engineers to deliver production features, resolve bugs, and improve overall user experience.

PROJECTS

Thai Election Dashboard

May 2026

Claude Code, Gemini OCR

- Developed an analytical dashboard to visualize and analyze Thai election data from Kalasin Province using Claude Code and Gemini OCR.
- Built interactive data visualizations to provide insights into voting patterns and election results.

Restaurant Reservation Website

Aug 2025 – Dec 2025

React Native, Express, PostgreSQL

- Developed a full-stack restaurant reservation platform using React Native, Express, and PostgreSQL.
- Designed the relational database schema and ER diagram for reservation management.
- Applied Agile development practices using GitHub and Jira for collaboration and project management.
- Gained hands-on experience in API design, backend development, database design, and teamwork.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, C, SQL

Frontend: Svelte 5, HTML, CSS, Bits UI

Backend: Node.js, Express, GraphQL

Database: PostgreSQL, MongoDB

Tools: Git

AWARDS

Silver Medal – Thailand Olympiad in Informatics 18

INTERESTS

Algorithms, Data Structures, Software Development, System Design, Artificial Intelligence

Ratsaporn Prathumma

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- github.com/RATSAPORN •

Computer Engineering student specializing in machine learning and artificial intelligence, with proven experience delivering end-to-end data pipelines and predictive models. Driven by a belief that well-engineered AI systems can meaningfully improve how people work and interact with technology, and eager to bring that perspective to challenging, real-world problems.

Education

Chulalongkorn University

B. Eng. In Computer Engineering GPAX [3.47]

August 2023 – May 2027

Experience

Backend Developer at SiriSoft Public Company Limited

June 2026 - July 2026

- Built microservices for an auction platform and developed a proof of concept (POC) integrating the Kong API Gateway for authentication and request routing.
- Strengthened backend development skills, including PostgreSQL database management, Apache Kafka message queues, Docker containerization, and microservice architecture.

Project

Thai Election Dashboard

May 2026 - May 2026

- Developed an analytical dashboard to visualize and analyze Thai election data (Kalasin Province) using Claude Code and extracted data using the Gemini OCR.
- Built interactive data visualizations to provide insights into voting patterns and election results.

Image Captioning Model with VMamba

March 2026 - May 2026

- Designed and built an **encoder-decoder** image captioning model based on a **VMamba-Transformer** architecture, applying **deep learning** techniques to integrate the encoder and decoder
- Wrapped the model with **FastAPI** and built a **Streamlit** interface to enable live captioning, testing, and deployment

SonKanBanNoi Cloud Automation

April 2026 - April 2026

- Architected a serverless **AWS** application to automate homework notifications from a university website, leveraging Lambda, EventBridge, and OpenSearch
- Engineered an **automated web scraping data pipeline** to collect and extract assignment deadlines, course materials, and user data

Multilabel Text Classification with PhayathaiBERT

February 2026 – March 2026

- Built **Python** pipelines for preprocessing and evaluation, improving model performance through tuning and data handling

Chess Move Detection

October 2025 – December 2025

- Implemented a **U-Net model** for chessboard segmentation and a **YOLO-based model** for chess piece recognition and position detection, achieving **mAP@0.5–0.95 ≈ 0.811**, **precision ≈ 0.965**, and **recall ≈ 0.97**.

HewKawJang Reserving Restaurant Platform

August 2025 – December 2025

- Applied **Agile development** for teamwork management and utilized **GitHub** for collaboration in the progress of the project.
- Using **React Native** as a framework for Frontend development and **NodeJS** as a framework for backend development.

Activities

LINE HACK 2025

November 2025 – November 2025

- Developed a LINE Mini App using LINE APIs (LIFF, Messaging API) with **Next.js** for the frontend and **Express.js** for the backend.
- Integrated **AI functionality using the Gemini API**, focusing on effective prompt design to generate accurate and relevant responses.

Skills

Technical: JavaScript, Java, Python, Dart (Flutter), SQL, C++, TypeScript, React, AWS, Go

Language: English (Intermediate), Chinese (Basic understanding)